

Logical Reasoning Overview

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP) — **Lecturer:** Dr. Mohammad Abdulaziz

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1. Background & Functional Programming Revision

→ Last Video Revision:

- **Revise basic functional programming:**

- Currying ($f :: \alpha \rightarrow \beta \rightarrow \gamma$)
- Lambda expressions ($\lambda x. e$)
- Polymorphic types (type variables ‘a’, ‘b’)

→ Basic Isabelle Interface:

- **Manipulating theory file (.thy)**
- **Writing terms:**
 - Numbers (nat, int, real)
 - Lists ($[], \#, @$)
 - Sets ($\{x. P\ x\}, \cup, \cap$)
 - Products / tuples ((a, b))
 - Functions ($f :: \alpha \rightarrow \beta$)
- **Type annotations** for all these terms ($t :: \text{type}$)

2. Logical Specifications & Isabelle Proof Overview

→ How to Write Down a Logical Spec:

- **Logical Connectives:**

- Implication ($\longrightarrow, \implies$)
- Conjunction ($\wedge$), Disjunction ($\vee$), Negation ($\neg$)
- Quantifiers ($\forall, \exists$)

- **How do we write theorems** (lemma, theorem)

→ Performing Proofs in Isabelle:

- **Logical Reasoning:**

- Forward reasoning (FW)
- Backward reasoning (BW)

- **Substitution**

- **Mathematical Induction** (on natural numbers nat)